

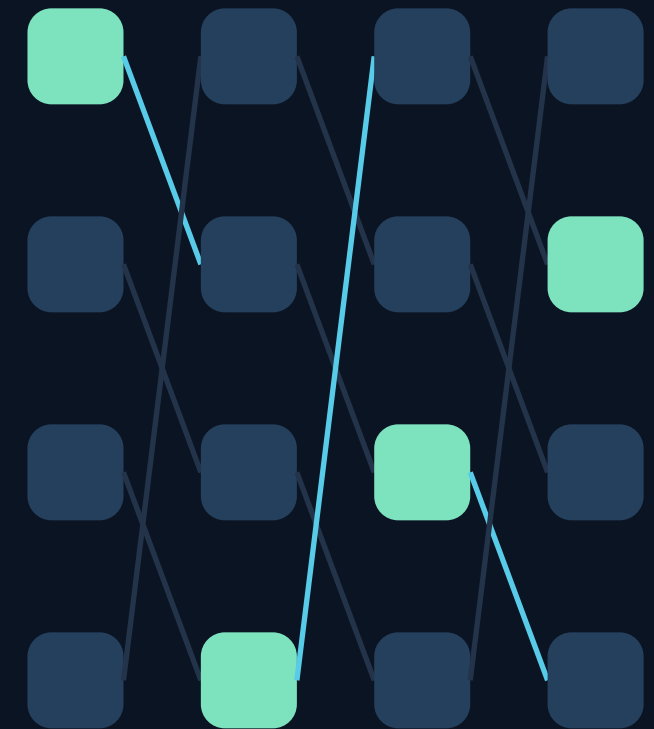
# Sleeping Machines

The next general-purpose substrate for machine intelligence.

One learner. Many forms of experience. Reusable intelligence, from representations to execution.

**Tero Keski-Valkama | Sole founder**

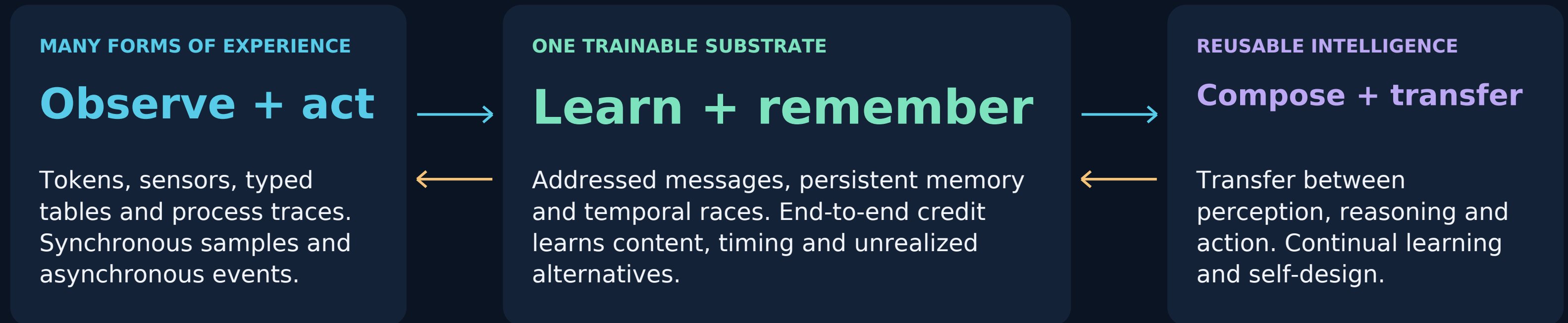
Raising €3M to scale research and engineering  
Research-stage deep tech · 4 October 2026



**Capacity beyond activity**

# One substrate for learning, representing and executing intelligence.

Different forms of experience can teach reusable computations within the same learner.



**The platform thesis: knowledge learned from one form of experience can become useful computation in another.**

Time and sleeps perform computation. Target: clockless learning and inference, capacity beyond activity, small data to large scale.

**First bridge: mixed-type tables through the temporal core. First transfer proof: joint learning, retained skills, matched work.**

# Public-leaderboard wins: timed event data (EasyTPP).

Public benchmark (ICLR 2024), official splits. Log-likelihood per event in nats: higher is better. Sealed test, 5 seeds.

| Dataset                         | Best published | Ours           | Result and compute we used                              |
|---------------------------------|----------------|----------------|---------------------------------------------------------|
| Taobao (shopping)               | 1.318          | 1.399 ± 0.003  | Win · +0.081 nats/event at 0.92× the leader’s compute   |
| Taxi (mobility), 5-seed mixture | 0.522          | 0.536          | Win · beats every published model at 0.41× its compute  |
| Taxi, single model              | 0.522 ± 0.004  | 0.525 ± 0.001  | Ahead on the mean at 1/12 of its parameters and compute |
| StackOverflow (Q&A activity)    | −2.163 ± 0.009 | −2.144 ± 0.004 | Win · +0.019 nats/event at 1.26× the leader’s compute   |
| StackOverflow, matched size     | −2.163 ± 0.009 | −2.153 ± 0.005 | Win · all 5 seeds ahead at 1.0× the leader’s compute    |
| Amazon (shopping reviews)       | 0.781 ± 0.011  | 0.803 ± 0.001  | Win · all 5 seeds ahead at 0.29× the leader’s compute   |
| Retweet (social)                | −6.348         | −6.326 ± 0.001 | Win at 1/15 of S2P2's compute; all 5 seeds ahead        |

**All five datasets won, each at about the state of the art’s compute or far less.**

Leaders: S2P2 (NeurIPS 2025) on Taxi, StackOverflow, Amazon; IFTPP on Taobao; NHP on Retweet. Compute = per-event inference multiply-adds.

# Public-benchmark win: ICU sepsis prediction (P19).

PhysioNet 2019, 38,803 ICU stays, 34 irregularly sampled channels, 4.2% positive. Five official splits, TEST once per split.

| Model                             | AUROC         | AUPRC         | Result                                  |
|-----------------------------------|---------------|---------------|-----------------------------------------|
| Best published (MTM, 2025)        | 0.903 ± 0.020 | 0.583 ± 0.053 | State of the art on the official splits |
| Previous leaders (ViTST, GraFITi) | 0.892 / 0.891 | 0.531 / 0.566 | Published                               |
| Ours, 62,681 parameters           | 0.916 ± 0.022 | 0.639 ± 0.039 | Win · AUPRC +0.056, beyond split spread |

The same temporal memory that won the event benchmarks wins clinical early warning.

AUPRC is the clinically relevant metric at 4% prevalence. AUROC lead (+0.013) is within one split sd.

# Beats the saved Transformer baselines at a fraction of the compute.

Wikipedia text (text8); same data and test set for all. Error: bits/char (bpc) or sMAPE %; lower is better.

| Comparison                                     | Their error       | Our error | Result and compute we used                                 |
|------------------------------------------------|-------------------|-----------|------------------------------------------------------------|
| vs Transformer, 4 layers, 4 passes (10M chars) | 1.908 bpc         | 1.888 bpc | Better · 40% of its training, 18% of its inference compute |
| vs Transformer, 2 layers, 1 pass (10M chars)   | 2.427 bpc         | 1.955 bpc | Better · 96% of its training, 16% of its inference compute |
| vs LSTM-256 (10M chars)                        | 2.171 bpc         | 1.955 bpc | Better at 88% of its inference compute (more training)     |
| vs LSTM-512, 6 passes (10M chars)              | 1.799 bpc         | 1.888 bpc | Behind by 0.09; a wider model did not close it             |
| vs best tuned LSTM (6 tuned runs)              | 1.825 bpc         | 1.888 bpc | Behind by 0.06; it used 92% of our compute                 |
| vs best tuned Transformer (4 tuned runs)       | 1.996 bpc         | 1.888 bpc | Better by 0.11 at ~equal compute (102%)                    |
| vs tuned Transformers (90M chars, ~2 PF)       | 1.811 / 1.704 bpc | 1.783 bpc | Better than TF-256×4 at 1.06×; behind TF-192×4             |

**Wins vs both saved Transformers; a tuned small LSTM still leads. That gap is the next proof.**

Compute = estimated arithmetic operations. Baselines trained by us; 10 tuned runs at our two budgets complete, selected by validation. Scoreboard: evidence pack R1.

# Event-driven models aimed for parity. Ours beats small Transformers on compute.

Published event-driven and spiking language models, as reported by their authors, beside our completed result.

| Model                                       | How it computes                                                              | Reported language result                                                          |
|---------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| EGRU on SpiNNaker2 (2023)                   | Clocked recurrent net; units fire events above a threshold                   | Matches LSTMs                                                                     |
| SpikeGPT, 216M (2023)                       | Clocked binary spikes in a recurrent Transformer variant                     | “Competitive” with non-spiking models                                             |
| Loihi 2 LLM, 370M (2025); sparse SSM (2026) | Conventional models made sparse/quantized; the chip supplies asynchrony      | No accuracy loss / comparable to dense                                            |
| Timing-coded LLM, 1.5B (2026)               | Spike times carry values; closest to our races                               | “Clear gap” in language perplexity                                                |
| Sleeping Machines (ours)                    | Learned arrival times choose which memories update; unchosen paths learn too | Beats two small Transformers at 40% of training compute; tuned LSTMs lead by 0.06 |

**To our knowledge, the first event-timed model to beat Transformer baselines on language with less compute, so far at small scale.**

Different datasets and scales; not a ranking. Ours: text8, 10M characters, small models, single runs, our baselines; LSTM-512 still ahead.



# Intelligence must compose across experience and execution.

A common substrate connects what a model learns with how and where it computes.

## REPRESENTATION

### Meaningful inputs

Tokens, sensor events and typed comparisons enter through appropriate interfaces and act on shared persistent state.

## LEARNING

### Reusable skills

End-to-end credit connects perception, memory, reasoning and action. Joint training tests transfer and retention.

## EXECUTION

### Adaptive work

Learn timing and selective activity; recruit useful capacity for the task and the resources available.

**Small-data tables and large-data language are complementary tests of the same platform thesis.**

# One model family across today's architectural landscape.

Our approach combines strengths of today's main model types.

## Recurrent networks / Mamba

Exact member: learned delays over decaying memory (checked).

## Transformers

Exact member: races sample the softmax; delays give attention.

## Mixture of experts

Activate part of a large model.  
We add private memory and timing.

## SLEEPING MACHINES / STATEFUL TEMPORAL PROGRAMS

One interface for content, timing, memory address and state, from single units to complete systems.  
Target: computation through timing, sparse memory updates, small messages, and learning from unchosen paths.

## Statistics / retrieval

Counts and lookups of past data.  
Combined with learned features.

## Adaptive memory

Models that adapt while in use.  
With local, causal updates.

## Event-driven systems

Event-driven, neuromorphic systems.  
We add trainable messages.

Mamba and attention containments are exact identities with numerical checks; other relationships are design relationships to be measured.



# Choose the right computation in each region.

One family can mix selective temporal programs with rich interaction and causal memory.

## 01 / UNITS

### Local programs

Each unit chooses what to remember, what message to send and when. Timing itself performs computation.



## 02 / LAYERS

### Adaptive work

A layer can activate one message or many: dense where needed, lean elsewhere.



## 03 / COMPOSITIONS

### Shared evidence

Learned features, counts and stored history can all inform the same prediction.



## 04 / SYSTEMS

### Same learner

The same learning rule works in batches or on a live stream, including learning after deployment.

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**Counterfactual credit: learn from selected and unrealized routes**

**No global clock is required. Combined models, learning-after-deployment gains and chip energy are still milestones.**

# Messages race. Selected memories update. Alternatives learn.

A simple event loop connects representation, timing and sparse computation.

## 01 / ENCODE

### Signal + delay

A learned function produces a message and its arrival time from input and local memory.



## 02 / SELECT

### Temporal race

Arrival times select a receiver. Separate address keys and delivered values support sparse execution.



## 03 / REMEMBER

### Local state

Update the selected memory and propagate a new message into deeper computation.



## 04 / LEARN

### Alternatives

Use errors to improve message content and timing, including paths that were not selected.

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**Counterfactual credit: learn from selected and unrealized routes**

**Implemented and trained today in ordinary software (CPU); no special hardware needed to run it.**

# One family wins across different kinds of data.

One construction: memories that evolve with elapsed time, sparse addressed writes, typed comparisons, races. The P19 classifier reuses the event model’s temporal memory layer, same code and size.

| Data type                           | Benchmark                             | Result                                                     |
|-------------------------------------|---------------------------------------|------------------------------------------------------------|
| Generative event streams            | EasyTPP (5 public datasets)           | Win on all 5: Taxi, Taobao, StackOverflow, Retweet, Amazon |
| One shared core, four event domains | EasyTPP, joint training (development) | One memory matches or beats separate models on 3 of 4      |
| Irregular clinical records          | P19 sepsis, official splits           | Win: AUPRC 0.639 vs 0.583                                  |
| Wearable sensors (activity)         | PAM, official splits                  | Win: accuracy 0.978 vs 0.975 (4 of 5 splits ahead)         |
| Anonymous interleaved process logs  | FAS v1, 3 seeds                       | Win vs six generic controls (0.592 vs 0.559)               |
| Character language, 10M             | text8 vs tuned baselines              | Win vs Transformers at $\leq$ equal compute; LSTM leads    |
| Temporal reasoning, few examples    | Structured event tasks                | 99.7% vs Transformers 33-41%                               |
| Mixed-type tables                   | Synthetic; banknote                   | Learns the interaction; trees lead on real data            |

**S2P2, the event-benchmark leader, is shown on one task type. We win on several.**

Large-scale language (90M characters) and real tables remain behind; pretraining on one clinical cohort did not reliably transfer to another.

# Respect types. Race decisions. Compose learned state.

Meaningful comparisons become events before evidence enters a common neural representation.

## RESPECT THE FIELD

### Typed comparisons

Compare a quantity with a threshold in its units. Test category membership, Boolean conditions and missingness explicitly. Category codes need no invented distance or order.

## LEARN THE DECISION

### Temporal races

Comparison outcomes feed computational races and selected memory writes. Counterfactual consequences teach routes. Learned thresholds require branch, timing and factual-path credit; that extension is the next test.

## COMPOSE THE EVIDENCE

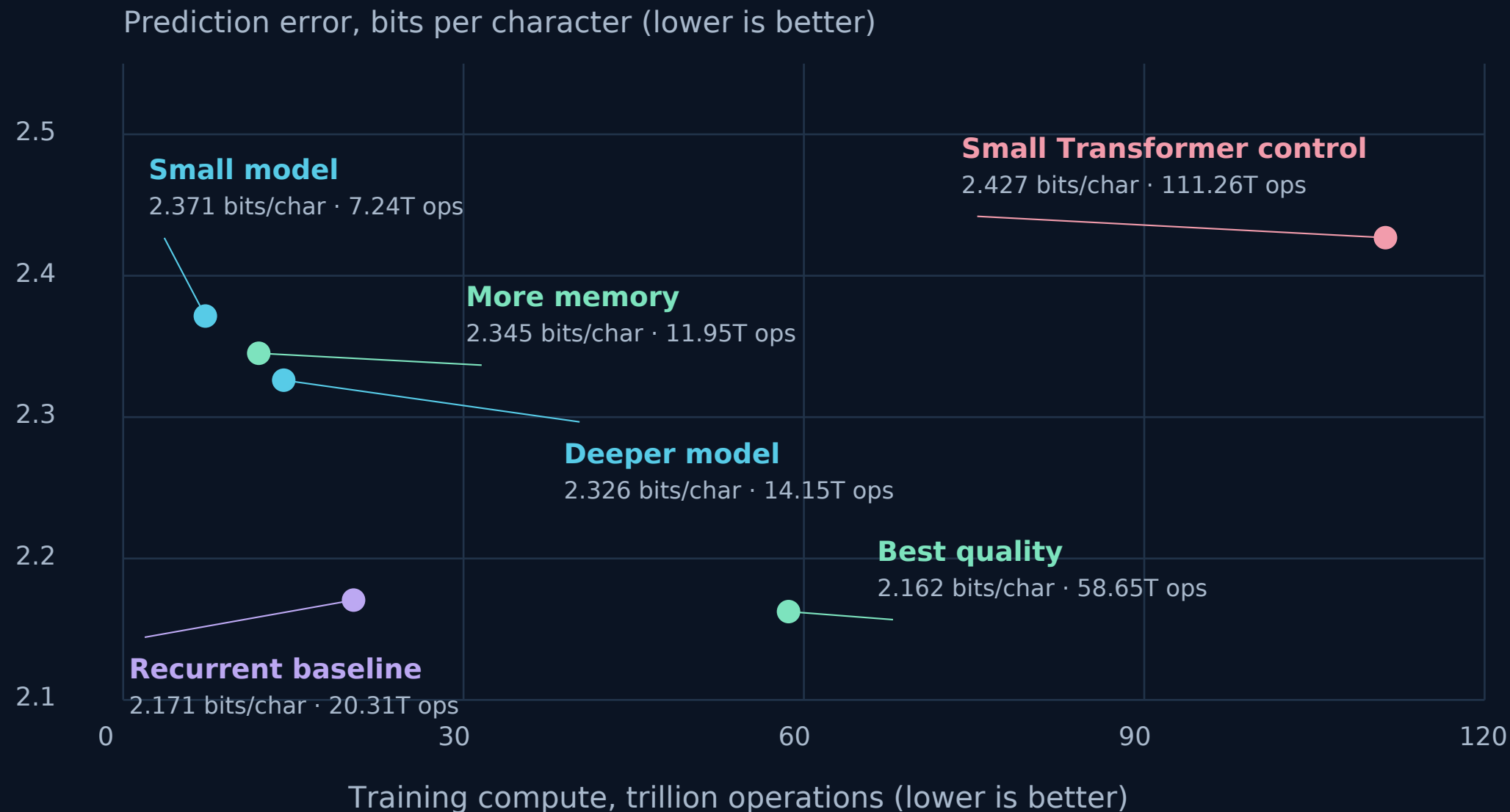
### Deep shared state

After a typed interface, learned vectors can mix meaningfully in persistent representations. The same event construction can connect tables, tokens, process traces and actions.

**Typed semantics + conditional selection + deep temporal learning. First mixed-type fit:100% synthetic DEV accuracy; fixed predicates.**

# Single pass: error versus training compute.

Each model sees the 10M training characters once. Lower and further left is better. Slide 2 adds multi-pass runs.



## BEST SINGLE PASS

# 2.162 bpc

Edges LSTM-256 (2.171), with 2.9× its training compute.

## TRAINING COMPUTE

# 15.4× less

Small model vs 2-layer Transformer: less compute and lower error.

Baselines trained by us on the same data and test set. Compute is estimated arithmetic (trillion operations), not measured energy or cost. Single runs.

# Start with low-batch language inference on existing hardware.

Product hypothesis for local and on-premise text applications; no customer interest reported yet.

## WHO WOULD EVALUATE IT

### Platform teams

Teams deploying small language models under latency, memory or cost constraints. Demand validation is a next step.

## WHAT WE WOULD SUPPLY

### Model + runtime

A trained model, sparse inference runtime and a reproducible quality-and-cost evaluation on the intended workload.

## WHY THEY WOULD ADOPT

### Verified savings

Adoption depends on equivalent useful quality, service reliability and a measured reduction in complete deployment cost.

**Commercial gate: target  $\geq 20\%$  full-system cost reduction at matched quality on one selected workload.**



# Software proves the advantage. Hardware multiplies it.

One research program, with staged routes to deployment.

## 01 / MODELS + RUNTIME

### First product

Validate the language workload on existing hardware. License a model/runtime with auditable deployment economics.

## 02 / MEMORY-LOCAL ACCELERATORS

### 1.7–5.8× less data

Modeled per character: 1.7× fewer memory bytes than the best tuned Transformer and 5.8× fewer than the saved one, both beaten. 10× more memory adds 5% traffic. Tuned LSTMs: parity.

## 03 / CLOCKLESS SILICON

### Work only on events

No global clock: idle parts draw almost no power, answers arrive as soon as they are ready, and cool chips allow dense 3D stacking. Also the missing model for neuromorphic chips.

**The €3M round funds scale and a hardware cost model of the trained network; silicon follows measured proof.**

# A useful alternative could strengthen a hardware ecosystem.

Potential evaluation and licensing routes; no partner interest or relationship is claimed.

## AMD

### Model + platform

Its ~\$665M Silo AI acquisition and Liquid AI collaboration show the value of model/software capability on AMD hardware.

## INTEL

### Event execution

Hala Point and Loihi 2 provide a relevant research environment for expressive event models and continuous learning.

## GOOGLE

### Full-stack design

TPU model-hardware co-design creates a demanding evaluation venue for quality, memory and execution improvements.

**Partner proposition: demonstrate a differentiated workload advantage that improves platform adoption.**



# Price the runtime against verified customer value.

Illustrative annual license economics; assumptions to test with an evaluation partner.

| Per-customer assumption | Calculation                                        | Annual amount |
|-------------------------|----------------------------------------------------|---------------|
| Eligible workload cost  | Existing deployment cost we can actually influence | €1.00M        |
| Gross savings           | 20% complete-cost reduction at matched quality     | €200k         |
| Runtime license         | 20% of verified gross savings                      | €40k          |

Customer retains €160k/year before migration and other costs. At the same assumed price, 25 deployments would generate €1M annual license revenue.

**Validate pricing through matched quality, reliable operation and measured customer benefit.**

# AI research and software architecture in one founder.

Tero Keski-Valkama · Sole founder · Public career and invention record.

## Tero Keski-Valkama

### Sole founder

Experience spans applied ML, multimodal foundation models and software architecture. The round adds specialist capacity around the founder.

2012-2018

**Cybercom · Software Architect**

Historical role in public professional profile.

2018-2022

**HERE · Lead ML & AI Engineer**

Self-reported research and applied ML leadership.

2022-2025

**Alloy.ai → Kaiko.ai**

Staff engineering; multimodal foundation-model work.

2019-2025

**35 patent publications · Inventor**

US and EP patents; assignee HERE Global.

Historical roles are self-reported; inventorship is documented. Planned hiring: learning research, runtime engineering, evaluation and measurement.

# Build defensibility across the model and its execution.

The strongest potential asset is an integrated stack that is difficult to reproduce and deploy.

## MODEL + LEARNING

### Research know-how

Timing, state and alternative-credit methods, supported by retained ablations and numerical contracts.

## RUNTIME + HARDWARE

### Execution know-how

Sparse kernels, memory management, measurement and hardware mapping can reinforce the model advantage.

## RIGHTS + PATENTS

### Company assets

Secure company ownership. Assess technical inventions for selective patent filings; protect confidential implementation know-how.

**Funding builds company-owned IP: ownership review, selective patent filings and confidential know-how.**

# Three gates determine whether this becomes a business.

Each gate has a falsifiable result and a consequence for further spending.

## 01 / QUALITY AT SCALE

### Replicate

At least three runs per result, fixed held-out tests and tuned baselines. Compare quality and total compute; publish failures too.

## 02 / COMPLETE ECONOMICS

### Measure

Show the fast sparse runtime gives the trained model's answers. Measure serving cost, latency, memory and energy on one workload.

## 03 / ADOPTION

### Evaluate

Identify a use case and an evaluator willing to test it. Confirm useful quality, integration effort and willingness to pay.

**If a gate fails, diagnose or narrow the workload before committing more compute or new silicon.**

# The next 18 months produce evidence investors can underwrite.

Proposed milestones measured from funding; spending advances with completed gates.

## MONTHS 0-3

### Team + proofs

Hire the initial team.  
Resolve IP ownership and review priority filings.  
Verify the trained sparse runtime; repeat larger-data results.

## MONTHS 3-6

### One workload

Run tuned controls and measure total cost, latency, memory and energy. Select a workload only if quality and economics hold.

## MONTHS 6-12

### External trial

Seek independent reproduction and prospective customer evaluations. Test pricing, integration effort and willingness to pay.

## MONTHS 12-18

### Product + scale

Turn a validated workload into a deployable runtime. Expand models and assess hardware mapping for the next financing stage.

**Next-round objective: replicated quality, a measured cost advantage and a credible evaluation pathway.**

# €3M funds an 18-month research and engineering program.

Team, staged compute, hardware feasibility and IP protection, including selective patent filings.



- Team: research + systems **€1.35M | 45.0%**
- Compute + independent replication **€0.90M | 30.0%**
- Hardware feasibility + measurement **€0.25M | 8.3%**
- IP, legal, operations **€0.20M | 6.7%**
- Contingency reserve **€0.30M | 10.0%**

**€2.70M operating plan + €0.30M reserve · Planned operating burn averages €150k/month.**  
Budget assumptions: six average FTE including founder at €150k/year fully loaded; compute and hiring quotes remain to be obtained.



# Raise €3M to establish a commercially valuable platform.

Proposed priced round; terms remain subject to company formation and investor diligence.

## CAPITAL TO RAISE

**€3M**

18 months of research, systems engineering and staged validation.

## PROPOSED PRE-MONEY

**€50M**

A premium research-stage proposal for the model/runtime/substrate opportunity.

## INITIAL OWNERSHIP

**5.66%**

€53M post-money, before option-pool changes, preferences and later dilution.

The proposal prices a model/learning family, sparse runtime and future chip architecture that already beats Transformer baselines at a fraction of the compute. A confirmed 90M-character win and a NeuroBench first place would justify a higher next price (the €100M scenario in the appendix).

**Valuation sensitivities and conditional exit scenarios are in the appendix.**

# Build the model. Prove the economics. Open the hardware path.

€3M proposed raise · €50M proposed pre-money · Private investor discussion.

## €3 million

Already beating Transformer baselines on compute.  
This round tests whether the advantage scales.

- 01** Replicated language quality and learning mechanisms.
- 02** A deployable runtime with measured system economics.
- 03** External evaluation and a credible path to hardware.

Tero Keski-Valkama · Private investor diligence available by arrangement